# SIMATS ENGINEERING
## ASSESSMENT TOOL 2

**COURSE NAME:** Artificial Intelligence **COURSE CODE:** CSA17

---

### COURSE OUTCOME COVERED

***CO4:*** *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

*Assessment Tool Weightage: 50%*

---

**QUESTIONS: 4** **Total Marks: 40**
**Duration: 1 hour**

Annexure A

### ARTICLE REVIEW

---

### STUDENT DETAILS

Name: V. Rupashree

Register Number: _________________________

Programme: B.E. Biomedical Engineering (Batch 2023–2027)

Department: SIMATS Engineering, Thandalam, Chennai

Date of Submission: _________________________

### Code of Conduct:

*I ____________________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*

*I understand that any violation of academic integrity rules will result in disciplinary action.*

**Signature of the student:**

## TASK 1: ARTICLE SUMMARY

### *Task 1(a) – Citation and Problem Definition*

**Full Citation:**

Almosained, A. M., Alotaiby, T. N., Alqahtani, R. A., & Murayshid, H. S. (2026). Classification of Heart Sound Recordings (PCG) via Recurrence Plot-Derived Features and Machine Learning Techniques. Electronics, 15(3), 601. https://doi.org/10.3390/electronics15030601

Publisher / Indexing: MDPI (peer-reviewed, open access). Published 29 January 2026.

Domain / Application Area: Biomedical signal processing applied to cardiovascular diagnostics — specifically, automated screening of heart valve disease from phonocardiogram (PCG) auscultation signals.

ML Problem Addressed: A supervised classification problem framed in two forms. The binary task separates normal from abnormal heart sounds, while the multiclass task assigns each recording to one of five categories — normal (N), aortic stenosis (AS), mitral regurgitation (MR), mitral stenosis (MS), and mitral valve prolapse (MVP). The core challenge is to design a feature representation that captures nonlinear cardiac dynamics accurately enough for classification, while remaining compact enough for deployment on low-power screening hardware.

### *Task 1(b) – Objective and Research Gap*

The analysis of cardiac acoustic signals plays a critical role in the identification and diagnosis of cardiovascular pathologies. Conventional diagnostic methodologies rely heavily on expert auscultation, a resource-intensive approach that may be impractical in low-infrastructure or underserved clinical environments. While deep learning models have demonstrated efficacy in classifying heart sounds, their deployment is often constrained by demands for extensive datasets, significant computational resources, and limited interpretability. In response to these limitations, the authors propose an alternative framework grounded in feature-based machine learning, designed to enhance transparency and accessibility. Their methodology involves transforming phonocardiographic signals into visual representations — specifically recurrence plots — that capture temporal dynamics and periodic structures inherent in the sound waveforms. From these graphical outputs, quantitative descriptors are extracted to characterise underlying patterns. Dimensionality reduction techniques are subsequently applied to refine the feature set, improving computational efficiency without compromising discriminative power. The reduced features are then employed to train conventional classifiers, including random forest, support vector machine, decision tree, and k-nearest neighbours. Experimental results demonstrated classification accuracies exceeding 98% in multiclass classification and approximately 99% in binary classification, indicating that recurrence-based descriptors can effectively distinguish pathological and normal heart sounds while maintaining interpretability and computational efficiency.

## TASK 2: METHODOLOGY ANALYSIS

### *Task 2(a) – Technique and Dataset*

The investigation was conducted using a curated dataset comprising 1,000 heart sound recordings, categorised into five distinct classes: aortic stenosis, mitral regurgitation, mitral stenosis, mitral valve prolapse, and normal cardiac function. Prior to analysis, preprocessing steps were implemented to standardise the data: each recording was resampled to 8 kHz, converted to a mono-channel signal, standardised to three cardiac cycles, and screened to remove recordings with excessive noise. Recordings exhibiting excessive noise were excluded to ensure signal integrity. To evaluate the impact of temporal resolution on classification performance, the signals were further partitioned into segments of one-second and two-second durations.

The core analytical technique employed in the study is based on recurrence plot (RP) generation — a nonlinear time-series visualisation method that reveals recurring states within dynamic systems. The phonocardiographic signal is first embedded into a phase space, where each point represents a reconstructed state of the system at a given time. Pairwise Euclidean distances between these points are computed and thresholded to produce a binary matrix, which is rendered as a two-dimensional image. This recurrence plot highlights structural regularities and transient behaviours not readily discernible in the raw waveform.

From each recurrence image, thirteen statistical and topological features were extracted, capturing attributes such as determinism, laminarity, recurrence rate, and complexity measures derived from matrix texture. To optimise model input and reduce redundancy, feature selection or dimensionality reduction was applied, resulting in a refined subset of eight features for multiclass classification and four for binary discrimination tasks. These engineered features served as inputs to four supervised learning algorithms. Model evaluation was performed using five-fold cross-validation, and performance metrics were averaged across all folds to provide a reliable assessment of classification performance.

***Task 2(b) – CO4 Mapping, Strengths, and Limitations***

The methodological framework aligns closely with foundational concepts in machine learning theory, particularly those encompassing decision tree learning, statistical inference, and inductive reasoning. Both decision trees and random forests exemplify hierarchical, rule-based classification strategies rooted in recursive partitioning of feature space. Support vector machines, conversely, operate within the paradigm of margin-based statistical learning, seeking optimal hyperplanes for class separation. Collectively, these models employ inductive principles — learning generalisable patterns from labelled instances to classify previously unseen data — thereby situating the study within established domains of automated pattern recognition and predictive modelling.

**Methodological Strength:**

A principal strength of the research lies in its deliberate feature engineering and selection process. The authors demonstrate that a minimal set of recurrence-based descriptors can maintain high classification accuracy, indicating that the chosen features effectively encapsulate diagnostically relevant information. This parsimonious representation enhances model interpretability and reduces computational overhead, making the approach more feasible for deployment in resource-limited settings.

**Methodological Limitation:**

A notable limitation pertains to the insufficient documentation of parameters governing recurrence plot construction. Specifically, the criteria for phase-space embedding — such as time delay and embedding dimension — and the thresholding mechanism used to binarise distance matrices are not explicitly justified or reported. Given that these parameters directly influence the topological structure of the resulting plots and, consequently, the derived features, their omission undermines methodological reproducibility. Furthermore, the reported embedding dimension of one effectively removes the benefits of phase-space reconstruction, raising questions regarding whether the generated recurrence plots fully capture nonlinear system dynamics. A systematic sensitivity analysis of these parameters would have strengthened the validity and robustness of the findings.

## TASK 3: RESULTS AND CRITICAL EVALUATION

***Task 3(a) – Key Results and Comparative Tables***

The proposed classification framework, based on recurrence plots, was assessed using four machine learning models: Random Forest, Support Vector Machine (SVM), Decision Tree, and K-Nearest Neighbours (KNN). Evaluations were carried out for both multiclass and binary classification scenarios, utilising one-second and two-second phonocardiogram signal segments. Performance metrics — accuracy and F1-score — were computed as averages over five-fold cross-validation. The results reveal a consistent trend: extended signal durations correlate with enhanced classification outcomes across all models.

**Table 1. Multiclass classification performance (five classes, full 13-feature set).**

| Classifier | Acc. 1 s (%) | F1 1 s (%) | Acc. 2 s (%) | F1 2 s (%) |
|---|---|---|---|---|
| Random Forest | 93.80 | 93.81 | 97.89 | 97.88 |
| SVM | 92.70 | 92.71 | 95.44 | 95.45 |
| Decision Tree | 89.60 | 89.57 | 95.00 | 95.02 |
| KNN | 96.10 | 96.09 | 98.44 | 98.45 |

In the multiclass setting, K-Nearest Neighbours exhibited superior performance relative to the other classifiers. When analysing one-second segments, KNN attained an accuracy of 96.10%, rising to 98.44% with two-second segments. Random Forest followed closely, demonstrating an increase from 93.80% to 97.89% under the same conditions. In contrast, Support Vector Machine and Decision Tree yielded lower classification accuracies, though both showed measurable gains with prolonged input duration. These findings imply that two-second recordings encapsulate richer temporal dynamics, thereby facilitating more effective feature discrimination among distinct cardiac pathologies.

**Table 2. Binary classification performance (normal vs. abnormal, full 13-feature set).**

| Classifier | Acc. 1 s (%) | F1 1 s (%) | Acc. 2 s (%) | F1 2 s (%) |
|---|---|---|---|---|
| Random Forest | 99.00 | 99.00 | 99.25 | 99.25 |
| SVM | 98.45 | 98.75 | 99.50 | 99.50 |
| Decision Tree | 97.25 | 97.24 | 97.50 | 97.50 |
| KNN | 99.00 | 99.00 | 99.50 | 99.50 |

For binary classification, all models achieved markedly high performance levels. SVM and KNN attained peak accuracy — approximately 99.50% — when processing two-second segments. Random Forest performed comparably well, reaching 99.25% accuracy, whereas the Decision Tree model recorded the lowest result at around 97.50%. The uniformly high performance across classifiers underscores the efficacy of features extracted from recurrence plots in reliably differentiating normal from abnormal heart sounds.

Table 3. Best configurations after NCA-based feature reduction.

| Task (2 s segments) | Best Model | Features Used | Accuracy (%) |
|---|---|---|---|
| Binary | SVM | 4 of 13 | $99.75 \pm 0.56$ |
| Multiclass (5-class) | KNN | 8 of 13 | $98.89 \pm 0.79$ |

A critical finding emerging from the results is the marginal gain in performance attributable to feature reduction. This suggests that the retained subset of recurrence quantification analysis (RQA) features preserves the majority of discriminative information present in the full set. Consequently, Neighborhood Component Analysis proves to be an effective dimensionality reduction strategy, substantiating the assertion that computational efficiency can be enhanced without compromising classification integrity.

***Task 3(b) – Critical Evaluation***

**Data Leakage Concern:**

A methodological limitation arises from the potential for data leakage during model evaluation. Although the study employed five-fold cross-validation on segmented recordings, insufficient detail is provided regarding whether segments derived from the same original recording were maintained within distinct training and validation folds. If temporally or physiologically related segments are distributed across both sets, the classifier may exploit recording-specific artefacts rather than learning generalisable pathophysiological patterns. This risk undermines the validity of the reported high accuracy rates, which may reflect over-optimistic performance estimates.

**One-Second versus Two-Second Segments:**

The authors observe superior classification performance with two-second segments compared to one-second intervals. While increased segment duration likely captures more complete cardiac cycles and enhances the stability of recurrence quantification features, this improvement may not be attributable solely to greater informational content. Variations in boundary alignment, noise distribution, or feature sensitivity due to differing window lengths could also contribute to performance discrepancies. Thus, the inference that longer segments are universally preferable requires further empirical justification.

**Lack of Independent Testing:**

The absence of an external validation cohort constitutes a significant constraint on the study's conclusions. Reliance exclusively on cross-validation within the same dataset limits the ability to assess true generalisability. Performance metrics obtained under such conditions may not reliably predict behaviour on independent, real-world data. Validation using demographically and clinically diverse external datasets would strengthen claims about the model's robustness and clinical applicability.

**Performance Metrics:**

The evaluation framework centres primarily on aggregate metrics — accuracy and F1-score — which may obscure critical aspects of classifier behaviour in imbalanced medical contexts. The omission of ROC-AUC and detailed class-wise performance measures restricts the depth of assessment. Although confusion matrices were reported, a more comprehensive presentation of per-class sensitivity and specificity would have strengthened the evaluation. Given the asymmetric clinical consequences of misclassification — particularly the risks associated with false negatives in cardiac screening — a more granular analytical approach is necessary to evaluate diagnostic reliability.

**Interpretability:**

While the authors assert that recurrence-based features offer physiological interpretability, the linkage between feature importance and underlying cardiac dynamics remains inadequately substantiated. Furthermore, several features highlighted in the physiological discussion, such as determinism and laminarity, were not among the most influential features identified during NCA-based selection, creating a gap between the interpretability claims and the reported feature importance results. A more rigorous analysis correlating dominant features with known haemodynamic phenomena would be required to support claims of meaningful interpretability beyond algorithmic efficacy.

**Additional Experiments That Would Strengthen the Findings:**

Three additions would materially improve confidence in the results. First, a recording-level (subject-independent) cross-validation protocol would demonstrate that the reported accuracy is not inflated by segment leakage. Second, a sensitivity study sweeping the embedding dimension, time delay, and recurrence threshold would establish how stable the RQA features are with respect to parameter choice. Third, controlled noise-injection experiments — adding respiratory sound, ambient noise, and motion artefact at graded signal-to-noise ratios — would quantify the degradation expected under realistic acquisition conditions.

***Task 3(c) – Real-World Applicability***

**Deployment Contexts:**

The proposed methodology holds potential for integration into digital auscultation platforms, remote health monitoring systems, and primary care screening tools. By enabling automated detection of abnormal heart sounds, the framework could assist clinicians in early diagnosis, particularly in resource-constrained settings where access to cardiologic expertise is limited. Its utility may be greatest in triage scenarios, where rapid preliminary assessment can guide referral decisions.

**Computational Challenges:**

Despite favourable classification outcomes, computational demands present barriers to deployment in low-resource environments. Because recurrence plot construction scales with the size of the signal representation, practical deployment on low-power embedded hardware may require downsampling strategies or computational approximations. Efficient implementation on embedded systems, such as portable digital stethoscopes, would necessitate algorithmic optimisation or hardware acceleration to meet latency and power consumption constraints.

**Data Availability Limitations:**

The experimental analysis was conducted on a controlled, publicly available dataset characterised by high signal quality and minimal artefacts. In contrast, clinical recordings frequently include background noise, respiratory interference, motion artefacts, and variability in acquisition equipment and operator technique. Consequently, the observed performance may degrade under less idealised conditions, highlighting the necessity of testing in uncurated, real-world environments before clinical adoption.

**Ethical Considerations:**

Deployment of automated diagnostic aids entails ethical responsibilities, particularly when erroneous outputs may influence patient management. False-negative results could lead to delayed intervention, while false positives may trigger unwarranted investigations and patient anxiety. These systems must therefore be designed and implemented as adjunctive tools, with clear boundaries defined between algorithmic output and clinical decision-making authority.

## TASK 4: REFLECTION AND LEARNING OUTCOME

### *Task 4(a) – Three Key Insights*

**Insight 1 — Feature representation can substitute for model complexity.**

This study underscores the critical role of feature engineering in determining machine learning outcomes. The effective transformation of raw phonocardiographic signals into recurrence plots and quantitative recurrence features enabled robust discrimination across classes, demonstrating that domain-informed representations can significantly enhance model performance even with standard classifiers. This connects directly to the inductive learning component of CO4: the hypothesis a learner can express is bounded by the representation it is given, so the quality of the input space determines how much generalisation the algorithm can achieve.

**Insight 2 — Ensembles reduce variance relative to single decision trees.**

The comparative advantage of Random Forest over single Decision Tree models illustrates the benefits of ensemble approaches in reducing variance and mitigating overfitting through aggregation of multiple weak learners. This observation reinforces the value of ensemble methods in achieving stable and reliable predictions, particularly in biomedical applications where data volume and diversity may be limited. It maps to the decision tree algorithms topic of CO4, extending the syllabus treatment of recursive partitioning and pruning to bagging-based ensembles, and is visible in the results as both a higher mean accuracy and a lower standard deviation across folds for Random Forest.

**Insight 3 — Evaluation design determines whether a reported result is meaningful.**

An essential insight pertains to the profound impact of evaluation design on research conclusions. The choice of validation strategy, data partitioning methodology, and performance metrics directly influences the interpretation of model effectiveness. This emphasises the need for meticulous scrutiny of experimental protocols in machine learning studies to distinguish genuine advances from artefacts of evaluation bias. Within CO4, this relates to model building and evaluation for classification: k-fold cross-validation, the bias–variance trade-off, and the selection of appropriate metrics for imbalanced or clinically asymmetric problems.

### *Task 4(b) – Proposed Extension*

**Proposed Experiment:**

The proposed extension is a strictly recording-independent revalidation of the framework, coupled with an embedded-hardware feasibility study. In this design, all segments originating from a single source recording would be confined to one fold, eliminating the leakage pathway identified in Task 3(b). The model trained under this protocol would then be evaluated on an independent public corpus, such as the PhysioNet/CinC 2016 heart sound database, which differs substantially in recording equipment, acquisition environment, and noise profile. In parallel, the reduced four-feature and eight-feature RQA pipelines would be ported to a microcontroller-class target and profiled for inference latency, memory footprint, and energy consumption per classification.

**Feasibility:**

The proposal is feasible using existing resources. Both the Yaseen 2018 dataset and the PhysioNet corpus are publicly available at no cost, and RQA feature extraction is implementable in Python or MATLAB without specialised hardware. The classifiers involved are standard library implementations. The embedded profiling stage requires only a low-cost development board with a microphone front end, which places the work within the scope of a final-year undergraduate project.

**Expected Impact:**

The expected outcome is a lower but far more credible accuracy figure. Cross-dataset performance would establish whether recurrence-based descriptors encode genuine pathophysiological structure or dataset-specific characteristics, which is the central open question left by the original study. The hardware profiling would convert the authors' efficiency claim from a qualitative assertion into measured latency and power figures, providing the evidence base needed before any point-of-care screening deployment could be justified.